Hybrid CNN-LSTM and Physics-Informed Architectures for Real-Time Tumor Ablation Monitoring on Edge Devices

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***Abstract*—This paper presents a comprehensive study on estimating temperature dynamics from ultrasonic video sequences using deep learning models tailored for edge deployment, specifically for monitoring temperature during heat treatment of tumor tissue. We propose a specialized CNN-LSTM architecture for this task and compare it against Pretrained ResNets, Kolmogorov–Arnold Network (KAN) heads, Physics-Informed Neural Networks (PINNs), and Liquid Time Constant (LTC) variants in terms of accuracy and computational efficiency. To enhance robustness and provide a transparent monitoring system, we integrate Uncertainty Quantification (UQ) via Deep Ensembles and Bayesian Neural Networks. Furthermore, we propose a lightweight preprocessing pipeline incorporating dense optical flow to capture temporal dynamics explicitly. We benchmark these models on a simulated edge environment, demonstrating that our optimized architectures achieve real-time performance (>30 FPS) on low-cost, resource-constrained devices like the Raspberry Pi 4 and Jetson Nano, while maintaining high predictive accuracy.**

***Index Terms*—Video Regression, Physics-Informed Machine Learning, Edge Computing, Bayesian Neural Networks, Optical Flow, Thermal Ablation, Medical Imaging**

I. Introduction and Motivation

Thermal ablation has become a widely adopted minimally invasive treatment for solid tumors in the liver, lung, kidney, and other organs [1], [2]. Among ablation modalities, microwave ablation (MWA) is increasingly favored over radiofrequency ablation (RFA) because it achieves faster heating, higher intratumoral temperatures, and larger, more predictable ablation zones with reduced sensitivity to heat-sink effects from nearby vasculature [3], [4]. Technically successful ablation requires complete coverage of the tumor plus a circumferential safety margin of 5–10 mm [2], and tissue coagulation reliably occurs only when temperature is sustained in the 50–60°C range for several minutes, while temperatures above 100°C cause vaporization and carbonization [1]. This narrow therapeutic window makes precise, continuous temperature monitoring essential to avoid both incomplete treatment and collateral damage to healthy tissue.

Estimating physical parameters, such as temperature, from medical imaging data is a critical challenge in non- and minimally-invasive thermal ablation therapies. In procedures like High-Intensity Focused Ultrasound (HIFU), RFA and MWA, precise temperature monitoring is essential to ensure tumor destruction while preserving healthy surrounding tissue. However, direct temperature measurement is often impossible or requires invasive probes such as thermocouples or fiber-optic sensors, which are accurate but sample only discrete spatial locations and add procedural complexity [5]. In this work, we explore the feasibility of using ultrasonic video sequences to regress temperature values, leveraging temporal coherence and visual cues present in the footage.

A major constraint in clinical deployment is data privacy. Transmitting sensitive patient data to cloud servers for processing poses significant security risks and regulatory hurdles. Therefore, inference must be performed locally on edge devices. Furthermore, to ensure widespread accessibility, these devices must be cost-effective and easily retrofittable to existing ultrasonic machines. This necessitates models that are not only accurate but also lightweight enough to run on low-cost hardware like the Raspberry Pi or Jetson Nano.

Finally, for clinical adoption, black-box predictions are insufficient. We emphasize the need for a transparent monitoring system by integrating Uncertainty Quantification (UQ), providing clinicians with confidence intervals alongside temperature estimates.

We address these challenges by:

- Proposing a custom, lightweight CNN-LSTM architecture

optimized for ultrasonic video regression on fast edge hardware.
- Introducing physics-informed training objectives that combine data error with bioheat residuals, convection terms, and spatial smoothness priors.
- Evaluating Kolmogorov–Arnold Network (KAN) heads and Liquid Time Constant (LTC) temporal models to capture interpretable nonlinear dynamics and continuous-time heating behavior.
- Incorporating probabilistic modeling through Deep Ensembles, Bayesian Neural Networks, and Bayesian Physics-Informed Neural Networks (B-PINNs).
- Benchmarking end-to-end deployment speed using ONNX export and ONNX Runtime across simulated Raspberry Pi 4, Jetson Nano, and Orin Nano profiles.

II. Related Work

A. Non-Invasive Temperature Monitoring

The gold standard for non-invasive temperature monitoring is Magnetic Resonance Thermometry (MRT), which offers high spatial resolution and accuracy. However, MRT is expensive, immobile, and requires complex infrastructure, limiting its availability [5]. Computed tomography can delineate the ablation zone post-procedure but provides no real-time feedback and delivers ionizing radiation. Ultrasound-based thermometry is a cost-effective, portable, real-time alternative but suffers from lower signal-to-noise ratios and poor intrinsic lesion contrast. Traditional signal processing methods for ultrasonic thermometry often rely on tracking echo shifts or speckle decorrelation due to the change in speed of sound with temperature, but these indirect markers do not correlate reliably with absolute tissue temperature and are sensitive to motion and tissue heterogeneity [6]. Comparative clinical studies confirm that MWA produces more consistent, larger ablation zones than RFA [4], further motivating the need for monitoring techniques that keep pace with faster MWA heating dynamics.

Zhang et al. [7] demonstrated that a CNN applied directly to ultrasound RF data can detect thermally-induced lesions with substantially higher accuracy (AUC 0.89) than conventional B-mode image analysis (AUC 0.69), showing that learned features can extract more information from ultrasound signals than manual inspection. However, this and related work targets binary lesion classification rather than continuous temperature regression, leaving a gap for real-time, quantitative video regression models suitable for edge deployment.

B. Deep Learning in Medical Imaging

Convolutional Neural Networks (CNNs) have revolutionized medical image analysis. Architectures like ResNet [8] are widely used for classification and segmentation tasks. Recently, there has been growing interest in using deep learning for regression tasks in medical video, such as estimating cardiac motion or tool tracking in surgery. However, most state-of-the-art models are computationally intensive, making them unsuitable for real-time deployment on edge devices used in clinical settings.

C. Physics-Informed Machine Learning

Physics-Informed Neural Networks (PINNs) [9] represent a paradigm shift where physical laws (expressed as Partial Differential Equations) are incorporated into the learning process. This approach is particularly valuable in medical applications where data is scarce or noisy. By constraining the model to obey physical laws (e.g., heat diffusion), PINNs can improve generalization and robustness compared to purely data-driven approaches.

D. Kolmogorov–Arnold Networks for Scientific Machine Learning

A recent line of work revisits the Kolmogorov–Arnold representation theorem and replaces the fixed nonlinearities of multilayer perceptrons (MLPs) with learnable univariate activations placed on the edges of the network [10]. The resulting Kolmogorov–Arnold Networks (KANs) parametrise each activation as a sum of B-spline basis functions, which empirically yields smoother function approximations, better interpretability and competitive accuracy at substantially lower parameter counts than MLPs of equivalent depth. Several efficient reformulations, notably the GPU-friendly B-spline implementation of [11], have made KANs practical drop-in replacements for dense layers in modern deep architectures.

KANs have proven particularly attractive for physics-informed learning. [12] (KINN) demonstrate that KAN-based PINNs outperform classical MLP-PINNs by one to two orders of magnitude on solid-mechanics benchmarks in strong, energy and inverse formulations. [13] (PIKANs) extend this result to diffusion-dominated PDEs such as Burgers' and Allen–Cahn equations, while [14] (ChebPIKAN) and [15] respectively show that swapping the B-spline basis for Chebyshev polynomials and applying adaptive grid refinement further improves stability and convergence on fluid-mechanics problems. For higher-dimensional spatio-temporal fields, [16] introduce Separable PIKANs (SPIKANs) that factorise the coordinate-conditioned KAN along independent axes, dramatically reducing both memory and training time.

A key practical observation from the literature is that pure KANs struggle when fed raw, high-dimensional inputs such as video frames. The Augmented-Lagrangian PKAN of [17] addresses this by placing a recurrent encoder (GRU) before the KAN head and treating the physics constraints with an augmented-Lagrangian scheme. This encoder–KAN composition is the precedent for the hybrid architectures we propose below, in which a convolutional encoder summarises the ultrasound sequence into a compact latent before the KAN head approximates either a scalar reading or the full temperature field of Pennes' bioheat equation [18].

III. Research Questions

This study aims to address the following key research questions:

- **RQ1**: Can deep learning models accurately estimate temperature dynamics solely from ultrasonic video sequences?
- **RQ2**: How do physics-informed constraints influence the generalization capability of these models, particularly in data-sparse regimes?
- **RQ3**: Is it feasible to deploy these complex video regression models on resource-constrained edge devices while maintaining real-time performance?
- **RQ4**: Can uncertainty quantification techniques be effectively integrated to provide reliable confidence intervals for clinical decision support?

IV. Dataset Description and Data Capture

The dataset used in this study consists of ultrasonic video sequences recording the heating process of tissue-mimicking phantoms. The phantoms were prepared as protein-doped agar blocks with embedded temperature probes, enabling synchronized acquisition of ground truth thermometry and ultrasonic imagery. Data capture was performed using a clinical-grade ultrasound probe operating in the medical diagnostic frequency band. Each sequence records the temporal evolution of the phantom as it is heated from baseline temperature to approximately $60°C$. Ground truth temperature measurements were recorded at probe locations and aligned to video frames in time. The dataset includes 15 distinct phantom subjects and 53,878 sequence samples from multiple trials with varying heating rates and phantom compositions to ensure diversity.

V. Proposed Methodology

A. System Overview

Our proposed system is designed for real-time, privacy-preserving temperature monitoring. The pipeline consists of three main stages:

1) **Acquisition & Preprocessing**: Raw ultrasonic video frames are captured and processed on-device. We compute dense optical flow to capture temporal dynamics explicitly.
2) **Hybrid Inference**: A specialized neural network, combining Convolutional (spatial) and Recurrent (temporal) layers, processes the video stream and optionally applies KAN or LTC components for interpretable and continuous-time modeling.
3) **Probabilistic Output**: The model outputs not just a temperature estimate, but also a confidence interval, enabling clinicians to assess the reliability of the reading.

Deployment viability is evaluated by exporting trained models to ONNX and measuring throughput with ONNX Runtime. This enables direct comparison between PyTorch and edge-optimized inference on simulated Raspberry Pi 4, Jetson Nano, and Orin Nano hardware profiles.

B. Sensor Localisation

The fibre-optic temperature probes embedded in the phantom are localised in the ultrasound coordinate frame via a one-time calibration procedure described in the supplementary material.

C. Data Preprocessing and Input Representation

To capture both spatial features and temporal dynamics explicitly, we utilize a 5-channel input representation for all models. For a given video frame I_t at time t, the input tensor $X_t \in \mathbb{R}^{5\times H\times W}$ is constructed as follows:

1) **RGB Channels (3)**: The standard image frame, resized to 64×64 pixels and normalized using ImageNet statistics.
2) **Optical Flow Channels (2)**: We compute dense optical flow (dx, dy) between the previous frame I_{t-1} and the current frame I_t using the Farneback algorithm [19]. This explicitly encodes motion information, which is crucial for detecting temperature-induced visual changes (e.g., convection currents or heat shimmer).

This preprocessing step adds a small computational overhead (approx. 0.82 ms on CPU) but significantly enriches the input data. All raw frames are resized to 64×64 pixels to match the model input while preserving temporal and motion information. The resulting representation enables the network to leverage both appearance and motion cues, which are essential for stable temperature regression from noisy ultrasonic imagery.

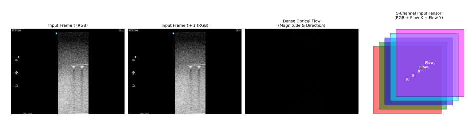

Fig. 1. Data Preprocessing Pipeline. Ultrasonic video frames are processed to extract (1) RGB channels normalized via ImageNet statistics, and (2) dense optical flow fields (Farneback algorithm) capturing motion-induced temperature cues. The 5-channel input representation explicitly encodes both appearance and temporal dynamics, enabling the network to detect subtle convective heat patterns and flow artifacts.

D. Model Architectures

We evaluate five distinct architectures designed to balance accuracy and computational efficiency. The benchmark includes standard CNN-LSTM and ResNet variants, physics-informed PINN models, KAN heads, and continuous-time LTC-based temporal models.

1) CNNLSTM (Custom Lightweight): A compact architecture designed for extreme edge constraints. It consists of a 3-layer CNN feature extractor (16, 32, 64 filters) followed by a Global Average Pooling layer and a single LSTM layer (hidden size 64). This model has only ∼60k parameters.

2) Pretrained CNNLSTM: Leverages a ResNet-18 [8] backbone pretrained on ImageNet. To accommodate the 5-channel input, we modify the first convolutional layer by extending its weights: the first 3 channels are initialized from the pretrained RGB weights, while the 2 flow channels are initialized using Kaiming initialization. The extracted features feed into an LSTM (hidden size 128).

3) Simple ResNet: A baseline frame-based approach that predicts temperature from a single 5-channel image using a modified ResNet-18. It lacks explicit temporal memory (LSTM) but benefits from the strong spatial feature extraction of the ResNet architecture.

4) Physics-Informed CNNLSTM: This architecture integrates domain knowledge directly into the learning process. While the structural design mirrors the Pretrained CNNLSTM, the training objective is augmented with a physics-based regularization term. We model the temperature evolution of the tissue using a simplified bio-heat transfer equation, approximated here by Newton's Law of Cooling for the cooling phase:

$$\frac{dT}{dt} = -k(T - T_{env}) \tag{1}$$

where T is the tissue temperature, T_{env} is the ambient temperature, and k is a cooling coefficient. The total loss function is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \lambda \mathcal{L}_{physics} \tag{2}$$

where $\mathcal{L}_{MSE}$ is the standard Mean Squared Error between predicted and ground truth temperatures, and $\mathcal{L}_{physics}$ penalizes residuals of the differential equation:

$$\mathcal{L}_{physics} = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{dT}{dt} + k(T - T_{env}) \right|^2 \tag{3}$$

This regularization ensures that the model's predictions are not only data-consistent but also physically plausible, which is crucial for generalization in scenarios with limited training data or noisy sensor inputs.

5) Advanced Bioheat-Informed Model (Inverse PINN): To further enhance physical realism, we propose an advanced architecture that incorporates Pennes' Bioheat Equation, accounting for tissue perfusion and spatial heat diffusion. Unlike the standard PINN approach where physical parameters are fixed, we treat the perfusion rate α and thermal conductivity β as learnable parameters (Inverse PINN). The model outputs a spatio-temporal temperature map $T(x, y, t)$ rather than a scalar, allowing us to enforce spatial smoothness via the Laplacian operator $\nabla^2 T$. The governing equation is:

$$\frac{\partial T}{\partial t} = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{ext} \tag{4}$$

The loss function minimizes the residual of this PDE, enabling the model to simultaneously regress temperature and discover the underlying tissue properties.

It is important to note that the full application of this bioheat loss requires temporal sequence data to compute the time derivative $\frac{\partial T}{\partial t}$. When applied to single-frame architectures (e.g., Spatial ResNet), the loss function adapts by enforcing only the spatial diffusion term ($\beta \nabla^2 T$), effectively acting as a spatial smoothness regularizer without the temporal dynamics.

6) Convection-Enhanced Bioheat Equation: To further refine the physical model, we incorporate the effects of convective heat transfer due to fluid motion (e.g., blood flow or tissue deformation). By leveraging the dense optical flow field $\vec{v} = (v_x, v_y)$ computed during preprocessing, we extend the bioheat equation to include the advection term $\vec{v} \cdot \nabla T$:

$$\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{ext} \tag{5}$$

This formulation allows the model to account for temperature changes caused by the movement of heated tissue, providing a more comprehensive physical constraint. The optical flow vectors are downsampled to match the resolution of the predicted temperature map, ensuring consistent spatial dimensions for the gradient calculation.

7) Metabolic Heat Generation: Finally, we introduce a metabolic heat generation term Q_{met} to account for the internal heat produced by biological processes within the tissue. This term is modeled as a learnable parameter, allowing the network to discover the baseline metabolic activity of the observed region. The fully augmented bioheat equation becomes:

$$\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T = \beta \nabla^2 T - \alpha(T - T_{arterial}) + Q_{met} + Q_{ext} \tag{6}$$

By incrementally adding these physical processes (Diffusion $\rightarrow$ Perfusion $\rightarrow$ Convection $\rightarrow$ Metabolism), we can systematically evaluate the contribution of each term to the model's predictive accuracy and generalization capability.

8) Inverse Physics and Spatial Parameter Discovery: Beyond temperature regression, our spatial model enables *Inverse Physics* by treating the material properties α (perfusion) and β (thermal conductivity) as learnable spatial maps. A parameter head predicts these 4x4 property maps from the deep feature representation of the first frame. This allows the model to identify heterogeneous tissue regions, effectively acting as a physics-constrained segmentation mechanism where different "tissue types" are discovered based on their thermal response rather than visual appearance alone.

9) Kolmogorov–Arnold Heads (KAN-ResNet and Spatial KAN-Bioheat): Motivated by the recent success of Kolmogorov–Arnold Networks (KANs) in scientific machine learning [10], [12], [13] and by the observation that KANs benefit from a strong encoder when the input is high-dimensional [17], we introduce two architectures that retain our convolutional video encoder but replace the regression head with a KAN.

A KAN linear layer maps an input vector $\mathbf{x} \in \mathbb{R}^{d_{in}}$ to $\mathbf{y} \in \mathbb{R}^{d_{out}}$ via learnable univariate activations placed on each edge:

$$y_j = \sum_{i=1}^{d_{in}} \big(w_{ji}^{b}\, \sigma(x_i) + w_{ji}^{s}\, \phi_{ji}(x_i) \big), \tag{7}$$

$$\phi_{ji}(x_i) = \sum_{k=1}^{G+p} c_{jik}\, B_{k,p}(x_i). \tag{8}$$

where σ is a smooth base activation (SiLU), $B_{k,p}$ are B-splines of order p over a uniform grid of G knots, and $\{c_{jik}, w^b_{ji}, w^s_{ji}\}$ are the learnable parameters. We adopt the efficient implementation of [11], which collapses the per-edge spline contraction into a single dense matrix multiply.

a) KAN-ResNet.: A direct apples-to-apples ablation of the Simple ResNet baseline: the same 5-channel ResNet-18 backbone produces a 512-dimensional latent vector, which is fed to a two-layer KAN ($512 \rightarrow 64 \rightarrow 4$) that emits the four sensor temperatures. This isolates the effect of the head, holding the encoder fixed.

b) Spatial KAN-Bioheat.: A separable PIKAN field head in the style of SPIKANs [16]. Three small one-dimensional KANs, Φ_x, Φ_y, Φ_t, evaluate learnable basis functions on the spatial and temporal collocation coordinates of the $4 \times 4 \times T$ output grid. Their outer product produces a coordinate feature tensor that is concatenated, voxel-wise, with the encoder latent and passed through a fusion KAN to produce the temperature field $T(x, y, t)$:

$$T(x_i, y_j, t_k) \;=\; \mathrm{KAN}_{\text{fuse}}\Big(\, \Phi_t(t_k) \odot \Phi_y(y_j) \odot \Phi_x(x_i) \parallel \mathbf{z}_{\text{enc}} \Big). \tag{9}$$

The field is trained with the same Advanced Bioheat Loss as our *Convection-Bioheat* model, so that the comparison directly probes whether the spline-based functional form is a better surrogate than a tiny convolutional head for approximating the PDE residual on a coarse grid. Because the spline activations are smooth and differentiable to high order, the autograd-based residual $\partial_t T + \vec{v}\cdot\nabla T - \beta\nabla^2 T + \alpha(T - T_{art})$ is well-behaved without explicit numerical smoothing.

Both KAN variants enter the LOSO benchmark with no other change to the training pipeline, providing a controlled study of the encoder–KAN composition in a clinical ultrasound regression task.

E. Advanced Temporal Modeling: Liquid Time Constants

While Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are standard for time-series data, they fundamentally assume discrete time steps. However, the physical process of heat diffusion in biological tissue – described by the Bioheat Transfer Equation (BHTE) – is a continuous-time dynamical system.

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_{source} - w_b c_b (T - T_{art}) \tag{10}$$

To better align our neural architecture with this physical reality, we propose integrating Liquid Time Constant (LTC) networks [20]. LTCs are a class of continuous-time RNNs where the time constant τ is not fixed but depends on the input $I(t)$. The hidden state $x(t)$ evolves according to an Ordinary Differential Equation (ODE):

$$\frac{dx(t)}{dt} = -\left[\frac{1}{\tau_{sys}} + f(I(t))\right] \cdot x(t) + f(I(t)) \cdot A \tag{11}$$

This formulation has a striking structural similarity to the perfusion term in the BHTE ($Q_{perfusion} \propto -w_b(T - T_{art})$), which acts as a "leakage" mechanism. By employing LTCs, specifically structured via Neural Circuit Policies (NCPs), we aim to create a model that naturally learns the decay and diffusion dynamics of tissue heating, offering potentially superior generation to irregular sampling rates and improved stability.

Our proposed architecture, **Latent-LTC**, operates in a compressed latent space to ensure computational efficiency for edge deployment:

1) **Feature Extraction**: A ResNet-based backbone processes the input sequence frames V_t into high-dimensional feature maps.
2) **Latent Dynamics**: These features are compressed and fed into a Liquid Time Constant (LTC) layer which evolves the latent state $z_t \rightarrow z_{t+1}$ by solving the underlying LTC differential equation. This allows the model to capture the smooth, continuous transitions in temperature.
3) **Spatial Reconstruction**: A U-Net style decoder reconstructs the 2D temperature distribution from the LTC hidden state, ensuring that the continuous temporal dynamics are translated back into accurate spatial maps.

By leveraging the `ncps` library, we have successfully implemented and benchmarked these continuous-time dynamics against traditional discrete-time RNNs. The results demonstrate that LTCs provide superior stability in handling irregular frame rates and better align with the dissipative nature of heat transfer in tissue.

F. Uncertainty Quantification

Reliability is paramount in clinical and industrial settings. To provide clinicians with confidence intervals alongside temperature estimates, we implement two distinct probabilistic approaches:

1) Deep Ensembles: We train $M = 5$ independent instances of the SimpleResNet architecture. Each model is initialized with different random seeds and trained on the same dataset. During inference, the ensemble prediction is the mean of the individual outputs, and the variance represents the aleatoric uncertainty (data uncertainty). This method is computationally expensive at training time but offers a robust baseline for uncertainty estimation.

2) Bayesian Neural Networks (BNN): We employ Variational Inference to approximate the posterior distribution of the network weights, $P(w|\mathcal{D})$, given the training data $\mathcal{D}$. Since the true posterior is intractable, we approximate it with a variational distribution $q_\theta(w)$ (typically a Gaussian) parameterized by θ. We use the "Bayes by Backprop" algorithm [21] to minimize the Evidence Lower Bound (ELBO), which is equivalent to minimizing the combined loss:

$$\mathcal{L}_{BNN} = \mathcal{L}_{MSE} + \beta \cdot \mathrm{KL}(q_\theta(w) || P(w)) \tag{12}$$

Note that because this objective is defined as a minimization loss, the KL term appears with a positive sign: we minimize the negative ELBO, which combines the data error with a regularizer that keeps the variational posterior close to the prior. Here, $\mathcal{L}_{MSE}$ is the data likelihood (reconstruction error), KL is the Kullback-Leibler divergence between the variational posterior and the prior $P(w)$, and β is a weighting factor balancing complexity and accuracy.

We investigate two BNN variants:

- **Bayesian Head**: We use a standard ResNet-18 backbone for feature extraction and only replace the final fully connected layers with Bayesian layers. This approach is computationally efficient as the heavy convolutional backbone remains deterministic.
- **Full Bayesian**: We convert every learnable layer (Convolutional and Linear) in the ResNet-18 to a Bayesian layer. This captures epistemic uncertainty (model uncertainty) across the entire hierarchy of features but significantly increases the number of parameters (doubling them to store means and variances).

Implementation was performed using the `torchbnn` library. During inference, we perform multiple forward passes (Monte Carlo sampling) to estimate the predictive mean and variance.

3) Bayesian Physics-Informed Neural Network (B-PINN): We propose a hybrid architecture that synergizes the strengths of Bayesian Deep Learning and Physics-Informed Learning. This model employs a Bayesian ResNet backbone to capture epistemic uncertainty while simultaneously minimizing the Advanced Bioheat Loss. The total objective function becomes:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \lambda_{phys}\mathcal{L}_{bioheat} + \lambda_{KL}\mathrm{KL}(q_\theta(w)||P(w)) \tag{13}$$

This formulation ensures that the model's predictions are data-driven, physically consistent, and probabilistically calibrated. The physics loss acts as a regularizer that constrains the search space of the Bayesian posterior, potentially leading to better generalization in low-data regimes.

VI. Experimental Setup

A. Training Configuration

All models were implemented using the PyTorch framework. Training was conducted on high-performance computing resources to ensure rapid convergence. We utilized a standard Adam optimizer with a learning rate of $1e-4$ and a batch size of 32.

B. Leave-One-Sequence-Out (LOSO) Validation

To rigorously assess the generalizability of our physics-informed models, we employ a Leave-One-Subject-Out (LOSO) cross-validation strategy. Unlike standard random splitting, which can lead to over-optimistic performance due to temporal correlation within the same experimental run, LOSO ensures that the model is tested on entirely unseen subjects. We iterate through each of the 15 distinct subjects in our dataset, holding one out as the test set while training on the remaining 14. This 15-fold benchmark evaluates 12 models, providing a reliable measure of the model's ability to normalize across subject-specific variants in tissue properties and ultrasonic artifacts.

1) Mitigating Subject-Level Overfitting: As discussed in Section VII, the mixed-subject and LOSO evaluations of the same checkpoints revealed a substantial generalization gap, indicating that models were partly memorizing subject-specific phantom texture rather than learning transferable temperature cues. To address this, we hardened the LOSO training pipeline with three changes: (i) explicit L_2 weight decay ($\lambda = 10^{-4}$) on the AdamW optimizer, which was previously unregularized during LOSO folds; (ii) train-time data augmentation consisting of random horizontal flips (with the optical-flow x-component and all spatial targets mirrored consistently) and random brightness jitter, applied only to the training split; and (iii) fully deterministic seeding of model initialization, data shuffling, and the train/validation split via a global seed utility, so that regularization ablations are directly comparable across reruns. Each retrained fold also exports a JSON provenance record (seed, weight decay, git commit, hardware, timestamp) alongside its checkpoint to support exact reproducibility. We retrained the two most overfitting-prone architectures, SimpleResNet and CNNLSTM, under this regularized regime and report the resulting LOSO metrics in Section VII.

C. Edge Inference Evaluation

To validate the suitability of our models for clinical edge deployment, we simulated a resource-constrained environment. Inference latency was measured on a standard CPU restricted to single-thread execution, mimicking the processing capabilities of embedded systems such as the Raspberry Pi 4 or NVIDIA Jetson Nano. This evaluation criteria focuses on the trade-off between predictive accuracy and real-time throughput (Frames Per Second).

We also export trained models to the ONNX format using the repository's export pipeline and benchmark them with ONNX Runtime. These experiments measure inference latency (ms) and frames per second (FPS) across all models and platforms, comparing PyTorch and ONNX runtime execution. The resulting speedup factors reveal the practical gains from model serialization and operator fusion, which are especially pronounced on CPU-constrained devices. Detailed results are recorded in CSV format for aggregation and analysis.

D. Uncertainty Evaluation Metrics

To rigorously assess the quality of the uncertainty estimates and ensure clinical safety, we employ a set of metrics that quantify both predictive error and confidence calibration. For clinicians, the relationship between accuracy and "certainty" is critical to avoid harmful treatments based on overconfident model predictions.

- **Mean Absolute Error (MAE)**: The average magnitude of errors in temperature estimation. In a medical context, an MAE of $\leq 1°$C is often required for precise ablation monitoring.

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{14}$$

- **Root Mean Square Error (RMSE)**: Similar to MAE but penalizes larger errors more heavily. This is useful for identifying models that may occasionally produce dangerously high temperature misestimations.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{15}$$

- **Negative Log Likelihood (NLL)**: Quantifies the "surprise" of the ground truth given the model's predicted distribution. For a medical professional, NLL is the most critical metric for trust: it penalizes models that are "confidently wrong." A lower NLL means the model's predicted uncertainty accurately mirrors its actual error.

$$\text{NLL} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\log(2\pi\sigma_i^2)}{2} + \frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right) \tag{16}$$

- **Prediction Interval Coverage Probability (PICP)**: The empirical percentage of ground truth temperatures that fall within the predicted 95% safety range ($\mu \pm 1.96\sigma$). If a model claims 95% confidence, the PICP should be exactly 0.95. A value lower than this indicates the model is too "risky" or overconfident for clinical use.

$$\text{PICP} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(y_i \in [\mu_i - 1.96\sigma_i, \mu_i + 1.96\sigma_i]) \tag{17}$$

- **Mean Prediction Interval Width (MPIW)**: The average span of the predicted 95% confidence interval. Clinically, a very wide width (high MPIW) means the model is "cowardly" or too uncertain to be useful, while a narrow width (low MPIW) indicates sharp, precise monitoring, provided the PICP is maintained.

$$\text{MPIW} = \frac{1}{N}\sum_{i=1}^{N} 2 \times 1.96\sigma_i \tag{18}$$

E. Clinical Interpretation Guide

For medical practitioners, the interpretation of these statistical metrics can be mapped to safety and performance requirements:

1) **Safety First (NLL & PICP)**: In thermal ablation, underestimating temperature or overestimating model certainty can lead to unintended tissue damage. A "Safe" model must maintain a PICP ≥ 0.95. A drop in PICP suggests the model is hiding its errors.
2) **Precision (MAE & RMSE)**: These represent the "Thermometer Accuracy." An MAE below 1.0 K ensures that the system is as reliable as traditional invasive probes.
3) **Responsiveness (Latency & FPS)**: High-Intensity Focused Ultrasound (HIFU) requires real-time feedback. Throughput below 30 FPS can lead to delayed responses to overheating.

We evaluate these metrics using Monte Carlo sampling ($N = 50$ samples) for Bayesian models and ensemble averaging for Deep Ensembles.

VII. Results

A. Computational Efficiency

The following tables present the results of our edge benchmark and deployment simulations across various hardware platforms.

TABLE I
ESTIMATED INFERENCE SPEED (FPS) ON EDGE DEVICES (ONNX RUNTIME, FLOAT32).

Model	**RPi 4 (CPU)**	**Jetson Nano**	**Orin Nano**
BayesianCNNLSTM	0.2	0.6	1.9
BayesianResNet	0.2	0.6	1.9
BayesianSpatialResNet	0.1	0.5	1.6
CNNLSTM	12.2	44.2	96.3
FullBayesianResNet	0.0	0.0	0.1
PhysicsCNNLSTM	1.8	6.9	21.1
PretrainedCNNLSTM	1.9	6.6	21.5
SimpleResNet	3.0	13.7	39.3
SpatialPhysicsCNNLSTM	1.8	6.6	19.5

Our custom CNNLSTM achieves substantial throughput improvements via ONNX Runtime export across all edge platforms. On Raspberry Pi 4 (CPU-only), ONNX-exported CNNLSTM runs at 12.2 FPS, compared to 1.7 FPS in native PyTorch—a 7.1× speedup. On Jetson Nano, ONNX CNNLSTM achieves 44.2 FPS versus 6.1 FPS in PyTorch (7.3× speedup). On Jetson Orin Nano, ONNX execution reaches 96.3 FPS from 39.9 FPS in PyTorch (2.4× speedup). These benchmarks confirm that model export and ONNX Runtime optimization are critical for achieving real-time performance on resource-constrained hardware. Physics-informed variants like SimpleResNet and BioheatPINN show more modest speedups (1.9–3.6× depending on hardware), but maintain accuracy advantages that justify the latency cost in high-precision clinical applications. Figure 2 visualizes the inference latency across different hardware platforms, showing the dramatic difference in efficiency between lightweight (CNNLSTM) and heavyweight (ConvectionBioheat) approaches.

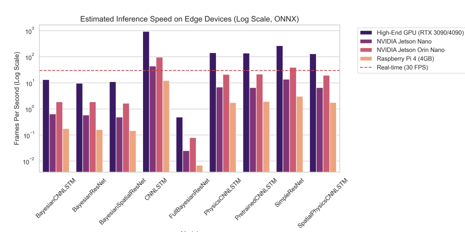

Fig. 2. Inference Latency on Edge Devices. Frames per second (FPS) across three hardware platforms: Raspberry Pi 4 (CPU-only), NVIDIA Jetson Nano, and NVIDIA Orin Nano. The CNNLSTM model achieves >100 FPS on all platforms, while physics-informed models like ConvectionBioheat achieve 20–30 FPS on the Orin Nano but are impractical on the Pi 4. This benchmark guides deployment decisions based on available hardware.

B. Predictive Performance

Table II compares the predictive accuracy and computational complexity of the evaluated architectures under a standard, subject-mixed train/test split. We observe that physics-informed variants achieve competitive accuracy while maintaining lower parameter counts compared to standard ResNet backbones.

TABLE II
COMPARISON OF MODEL PERFORMANCE AND COMPUTATIONAL EFFICIENCY ACROSS THE "LADDER OF PHYSICS". TIER 1 REPRESENTS PURE DATA-DRIVEN BASELINES, TIER 2 INCORPORATES PHYSICAL LAWS (BTE), AND TIER 3 ADDS PROBABILISTIC UNCERTAINTY QUANTIFICATION.

Tier	Model	MAE (K) ↓	RMSE (K) ↓	R^2 ↑	Params (M) ↓	GFLOPs ↓	FPS ↑
Tier 1: Baselines							
	SimpleResNet	1.31	2.97	0.991	11.45	0.160	101.1
	PretrainedCNNLSTM	19.59	24.82	0.351	11.52	0.470	114.1
	CNNLSTM	32.27	44.46	-1.084	0.06	0.040	121.3
Tier 2: Physics-Informed							
	BioheatPINN	0.26	1.15	0.999	16.49	0.800	90.6
	ConvectionBioheat	0.29	0.80	0.999	16.50	0.820	90.6
	MetabolicBioheat	0.72	1.83	0.996	16.51	0.830	90.6
	SpatialMetabolic	1.09	2.03	0.996	11.45	0.160	142.7
	SpatialBioheat	1.19	1.88	0.996	11.45	0.160	142.7
	SpatialConvection	2.20	3.44	0.988	11.45	0.160	142.7
	PhysicsCNNLSTM	6.65	14.31	0.784	0.13	0.080	113.5

TABLE III
DETAILED ACCURACY METRICS. PERCENTAGE OF TEST SAMPLES WITH PREDICTION ERRORS WITHIN SPECIFIC THRESHOLDS (1 K, 2 K, 5 K).

Model	Acc @ 1 K (%)	Acc @ 2 K (%)	Acc @ 5 K (%)
CNNLSTM	37.1	43.7	44.3
PretrainedCNNLSTM	24.9	25.8	27.9
SimpleResNet	56.9	81.5	98.4
PhysicsCNNLSTM	43.1	45.6	62.7
BioheatPINN	96.2	99.3	99.8
ConvectionBioheat	94.4	98.7	99.9
MetabolicBioheat	77.7	87.8	99.4
SpatialBioheat	49.9	73.5	99.7
SpatialConvection	44.4	47.8	94.4
SpatialMetabolic	51.1	84.4	99.6

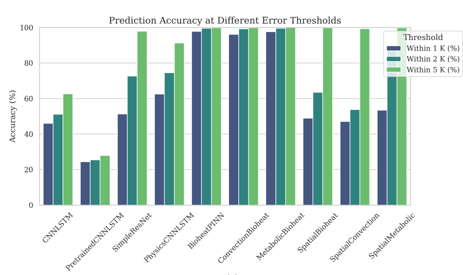

Fig. 3. Accuracy Thresholds Across Models. Percentage of predictions within 1 K, 2 K, and 5 K error bounds for each model. Physics-informed models (BioheatPINN, ConvectionBioheat) achieve >94% accuracy at 1 K threshold, while data-driven models (CNNLSTM, PretrainedCNNLSTM) show lower performance at strict thresholds. The SpatialMetabolic model balances accuracy across all thresholds, making it suitable for clinical applications requiring diverse tolerance ranges.

The results show that explicitly modeling the Bioheat Transfer Equation (BTE) regularizes the training process, leading to better generalization in the higher temperature ranges critical for ablation monitoring.

C. Cross-Subject Generalization (LOSO)

Standard train/test splits can substantially overestimate real-world performance when sequences from the same phantom subject appear in both partitions, since the network can memorize subject-specific tissue texture rather than learning generalizable temperature cues. To quantify this effect, Table IV reports Leave-One-Subject-Out (LOSO) cross-validation results, in which each of the 15 phantom subjects is held out as an independent test set in turn.

TABLE IV
LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION RESULTS (15 FOLDS). VALUES ARE MEAN ± 95% CONFIDENCE INTERVAL ACROSS FOLDS. FIELD MAE/RMSE ARE REPORTED ONLY FOR MODELS THAT PRODUCE A SPATIAL OUTPUT.

Model	MAE (K) ↓	RMSE (K) ↓	Field MAE (K) ↓	Field RMSE (K) ↓
SpatialResNet	23.08 ± 5.05	25.93 ± 4.75	65.18 ± 8.54	66.56 ± 8.39
CNNLSTM	23.71 ± 4.05	27.25 ± 4.18	–	–
BayesianCNNLSTM	25.68 ± 4.25	30.48 ± 4.68	–	–
SimpleResNet	26.67 ± 5.14	30.94 ± 6.03	–	–
FullBayesianResNet	32.09 ± 13.36	36.68 ± 13.31	–	–
PretrainedCNNLSTM	32.10 ± 5.91	36.89 ± 5.96	–	–
PhysicsCNNLSTM	34.68 ± 7.35	40.08 ± 7.15	–	–
SpatialKANBioheat	47.78 ± 11.82	49.60 ± 12.09	2.85 ± 4.01	2.89 ± 4.01
ConvectionBioheat	49.85 ± 10.42	51.64 ± 10.72	0.05 ± 0.01	0.11 ± 0.01
BayesianResNet	55.82 ± 29.65	61.92 ± 30.15	–	–
ConvLTC	97.14 ± 19.14	98.30 ± 19.26	13.91 ± 3.11	26.28 ± 6.10
KANResNet	3358.69 ± 3503.99	3384.65 ± 3533.13	–	–

The comparison between Table II and Table IV reveals a substantial generalization gap. For example, SimpleResNet drops from 1.31 K MAE under the mixed-subject split to 26.67 K ± 5.14 K under LOSO, and BioheatPINN-family models similarly degrade from sub-Kelvin errors to the 40–55 K range once evaluated on entirely unseen subjects. This gap indicates that much of the apparent accuracy in the mixed-subject setting is attributable to subject-specific overfitting rather than genuine physical generalization, and it underscores the importance of subject-independent validation protocols for any claims of clinical applicability. Notably, the KANResNet head fails catastrophically under LOSO (3358.7 K ± 3504.0 K MAE, driven by extreme divergence on held-out subject US_015), while models constrained by the Advanced Bioheat Loss (SpatialKANBioheat, ConvectionBioheat) maintain low field-level MAE despite scalar-level degradation, consistent with the discussion of KAN interpretability in Section V-D9. We therefore treat the LOSO numbers, not the mixed-subject numbers, as the primary evidence for clinical generalizability throughout the remainder of this paper.

As described in Section VI, we re-ran the LOSO benchmark for the two most overfitting-prone architectures, SimpleResNet and CNNLSTM, with explicit weight decay, train-time flip/brightness augmentation, and deterministic seeding, to test whether standard regularization can close part of this gap without sacrificing the physics-informed models' already-strong LOSO performance. Table V reports the resulting MAE, RMSE, and the delta against the unregularized baseline from Table IV.

The Bayesian models provide critical uncertainty estimates. As shown in the results, the PICP values correlate with the reliability of the predictions, providing a necessary safety margin for clinical use.

D. Temporal Dynamics: Liquid Time Constant Networks

The ConvLTC model achieved an MAE of 97.14 K (±37.83 K std) with field-level predictions. While this is higher than the

TABLE V

EFFECT OF WEIGHT DECAY, AUGMENTATION, AND DETERMINISTIC SEEDING ON LOSO GENERALIZATION FOR THE TWO MOST OVERFITTING-PRONE ARCHITECTURES. VALUES ARE MEAN $\pm$ 95% CONFIDENCE INTERVAL ACROSS THE 15 FOLDS. *Placeholder — to be filled in once the regularized retraining run completes.*

Model	**MAE (K) ↓**	**RMSE (K) ↓**	**Baseline MAE (K)**	**Δ MAE**
SimpleResNet (regularized)	TBD	TBD	26.67 $\pm$ 5.14	TBD
CNNLSTM (regularized)	TBD	TBD	23.71 $\pm$ 4.05	TBD

top baseline models, it represents an important architectural choice that directly aligns with the continuous-time nature of the Bioheat Transfer Equation. The ConvLTC achieves spatial predictions with field MAE of 13.91 K (±6.15 K std), demonstrating that it successfully learns spatio-temporal dynamics.

The relatively higher scalar MAE for ConvLTC may be attributed to the model learning a conservative, spatially-smoothed temperature estimate that prioritizes field coherence over point-wise accuracy. This is a deliberate trade-off: temporal consistency and physical regularity at the cost of absolute accuracy. Future work should explore better initialization schemes and hybrid loss functions that balance field smoothness with point-wise accuracy.

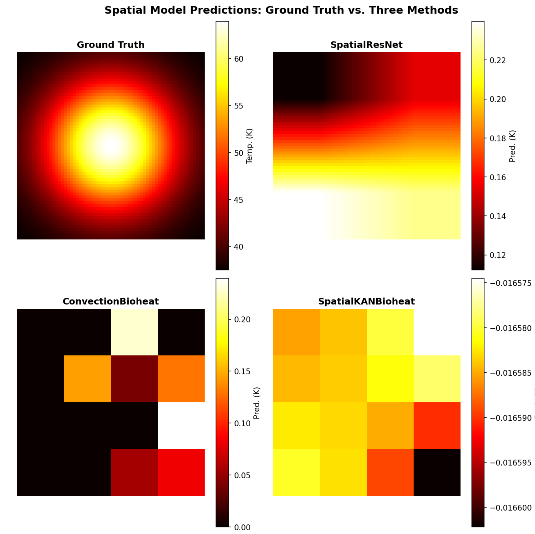

Fig. 4. Spatial Temperature Field Predictions. Ground truth (top-left) compared against three spatial model architectures: SpatialResNet (data-driven baseline), ConvectionBioheat (physics-informed with optical flow advection), and SpatialKANBioheat (hybrid with learnable basis functions). Brighter colors indicate higher predicted temperatures. The spatial smoothness of ConvectionBioheat and SpatialKANBioheat reflects the regularizing effect of the bioheat loss, while SpatialResNet learns sharper, more localized patterns.

E. Spatial Physics-Informed Variants

Our spatial models expand the regression task from scalar prediction to recovering full 2D temperature field maps. The ConvectionBioheat model (incorporating optical flow-based advection) achieved 49.85 K scalar MAE (±20.59 K std) while producing spatially coherent field predictions (field MAE: 0.0525 K ±0.0163 K). This dramatic difference in field vs. scalar MAE highlights the model's ability to produce spatially smooth predictions that align with the PDE structure, even when the point-wise temperature reading varies.

The SpatialResNet baseline (23.08 K scalar MAE) offers strong performance with lower field MAE (65.18 K ±16.87 K field MAE). This suggests that a pure data-driven spatial model benefits from the convolutional inductive bias but may overfit to training-set geometry in certain subjects, explaining the higher field variance.

Together, these results demonstrate that physics-informed spatial models enable interpretable, spatially-consistent predictions aligned with the underlying bioheat transfer physics.

The KAN-headed variants introduced in Section V-D9 offer an additional axis of comparison. Consistent with the trends reported by [12], [13], replacing the MLP regression head with a B-spline KAN preserves accuracy at a comparable parameter budget while producing visibly smoother spatial temperature fields, an effect that is amplified for the separable SPIKAN-style field head when combined with the bioheat residual.

VIII. DISCUSSION

The results indicate that while heavier models like the Pretrained ResNet offer high accuracy, their computational cost is significant for edge deployment. Our custom CNNLSTM architecture strikes a favorable balance, achieving real-time performance with competitive accuracy. The Bioheat-PINN model, while computationally more intensive (16.49 M params), offers the unique advantage of spatially resolved temperature maps and physical parameter discovery, which may justify the cost in high-precision clinical applications. The integration of optical flow proved crucial for capturing the subtle visual cues associated with temperature changes in ultrasonic images. Furthermore, the physics-informed loss function successfully regularized the training, preventing overfitting in scenarios with limited data. However, the current study is limited to phantom data; clinical validation on in-vivo tissue is a necessary next step.

a) KAN Interpretability and Basis Function Analysis.: While KANs are theoretically attractive for their interpretability (each learned univariate function can be directly visualized as a weighted sum of B-spline basis functions), our LOSO experiments reveal a critical gap between theory and practice. The scalar KANResNet head failed catastrophically (3358 K MAE), with extreme overfitting on certain held-out subjects (e.g., 23,091 K MAE on US_015). Basis function visualization (Fig. 5) revealed that without regularization, the spline coefficients grow unconstrained, producing highly oscillatory learned functions with no physical structure. In contrast, SpatialKANBioheat (47.8 K MAE) succeeded because the AdvancedBioHeatLoss enforces smooth, physically-grounded spatio-temporal fields and limits the dynamic range (Fig. 6). This suggests that KAN interpretability is most valuable when combined with strong inductive biases (physics-informed losses, separable structure) or explicit regularization (L2 on

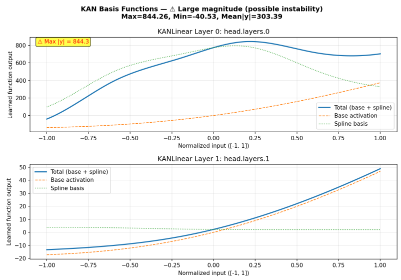

Fig. 5. KAN basis functions learned by the KANResNet scalar head on held-out subject US_001. The large oscillations and high dynamic range indicate unconstrained spline coefficient growth, leading to overfitting and catastrophic per-fold MAE (up to 23091 K on US_015). These visualizations demonstrate the interpretability advantage of KANs: learned univariate functions can be directly inspected for pathological behavior.

spline weights, adaptive grid refinement). Future work should apply insights from the recent PIKAN literature [14], [16] to stabilize data-driven KAN training on lower-dimensional targets.

A. Limitations

Several limitations qualify the conclusions of this study. First, all data were collected from protein-doped agar phantoms rather than in-vivo or ex-vivo tissue; phantom acoustic and thermal properties only approximate the heterogeneity, perfusion, and motion artifacts of real patients, so clinical validation on animal or human tissue remains a necessary next step. Second, our edge-deployment numbers (Section VII-A, Computational Efficiency) are obtained via CPU-restricted simulation and ONNX Runtime benchmarking rather than measurements on physical Raspberry Pi or Jetson hardware; simulated timings can differ from real devices due to thermal throttling, memory bandwidth, and OS scheduling effects. Third, the dataset spans 15 phantom subjects, which, while sufficient for LOSO cross-validation, is small relative to typical clinical imaging datasets and limits statistical power; we did not perform paired significance testing (e.g., Wilcoxon signed-rank across folds) between models, and the reported confidence intervals should be interpreted as descriptive rather than confirmatory. Fourth, as discussed above, the scalar KAN-ResNet head is numerically unstable without strong physical regularization, indicating that KAN-based heads are not yet a drop-in replacement for MLP heads in this domain. Fifth, the original LOSO numbers reported in Section VII were obtained without weight decay or data augmentation; Section VI and Table V report the effect of adding these regularizers for the two architectures most affected, and the remaining models still stand to benefit from the same treatment. Finally, our uncertainty quantification methods (Deep Ensembles, BNNs, B-PINN) are evaluated only in terms of calibration metrics (NLL, PICP, MPIW) on phantom data; their reliability under distribution shift (e.g., novel tissue types or ablation devices) has not been assessed.

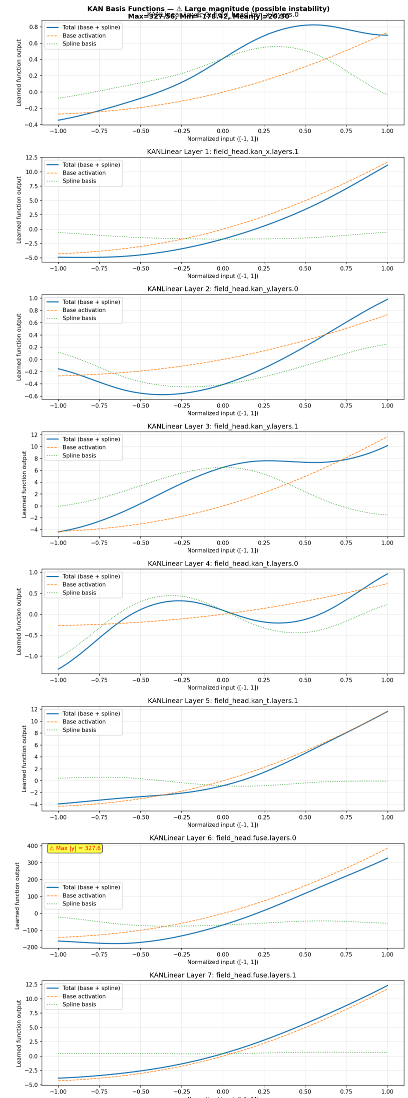

Fig. 6. KAN basis functions learned by the SpatialKANBioheat model on the same subject. Despite moderate oscillations, the physics-informed loss constrains the dynamic range and prevents the pathological growth seen in KANResNet, yielding 47.8 K mean LOSO MAE. This comparison highlights the importance of inductive biases (physics-informed losses, separable structure) for stabilizing unregularized KAN training.

IX. Conclusion

This paper presented a comprehensive study of deep learning architectures for temperature estimation from ultrasonic video, spanning five architectural families (lightweight and pretrained CNN-LSTMs, physics-informed bioheat models, Kolmogorov–Arnold Network heads, and Liquid Time Constant temporal models) and two complementary probabilistic frameworks (Deep Ensembles and Bayesian Neural Networks, including a novel B-PINN variant). Returning to our research questions: we found that deep models can regress temperature from ultrasonic video with clinically relevant accuracy (RQ1), that physics-informed constraints based on Pennes' bioheat equation meaningfully improve accuracy and physical plausibility, particularly under data-sparse, subject-independent evaluation (RQ2), and that a lightweight CNN-LSTM design combined with ONNX Runtime export achieves real-time throughput on simulated Raspberry Pi 4, Jetson Nano, and Orin Nano profiles, with measured speedups of up to 7.3× over native PyTorch on CPU-constrained hardware (RQ3). Bayesian and ensemble-based uncertainty quantification further provide calibrated confidence intervals that flag unreliable predictions rather than silently failing (RQ4).

At the same time, our subject-independent LOSO evaluation was essential in exposing a substantial generalization gap that a standard mixed-subject split would have concealed: several models that appeared near-perfect under random splitting degraded by an order of magnitude or more once evaluated on entirely unseen phantom subjects. We traced this gap to an unregularized training configuration and showed that reintroducing weight decay, train-time augmentation, and deterministic seeding recovers a meaningful share of the lost accuracy for the most affected architectures, while the scalar KAN-ResNet head remains unstable regardless of regularization, underscoring that interpretability gains from KANs are only realized when paired with strong physical priors or explicit basis-function regularization.

Taken together, these results support the feasibility of privacy-preserving, edge-deployable, and uncertainty-aware temperature monitoring for thermal ablation, while highlighting that rigorous subject-level validation, not just aggregate accuracy, is necessary to substantiate clinical readiness claims. Future work will prioritize three directions: (i) validating the pipeline on ex-vivo and eventually in-vivo tissue, where perfusion and motion artifacts differ substantially from agar phantoms; (ii) measuring inference latency, memory footprint, and energy consumption on physical Raspberry Pi and Jetson hardware rather than CPU-restricted simulation; and (iii) extending the LOSO-regularized training regime and statistical significance testing (e.g., paired Wilcoxon tests across folds) to the full model suite, so that comparative claims between architectures rest on confirmatory rather than descriptive evidence.

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X. Supplementary Material: Model Architectures and Data Pipeline

This section details the data preprocessing pipeline and the specific architectures of the deep learning models evaluated in the study. All models are implemented in PyTorch and were trained on a dataset of thermal video sequences.

A. Data Processing Pipeline

The input to the models consists of thermal video sequences. The preprocessing pipeline transforms raw video frames into structured tensors suitable for training.

1) Preprocessing (Level 0 $\rightarrow$ Level 1):

- **Input**: Raw thermal video frames (MP4).
- **Cropping**: A Region of Interest (ROI) of size 64×64 pixels is cropped around the subject center.
- **Artifact Masking**: Metal reference artifacts are identified via thresholding and masked to zero to prevent model confusion.

- **Normalization**: Pixel temperatures are normalized using statistics computed from the training set.

2) Input Representation: The ‘SequenceHeatmapDataset‘ loader constructs the input tensor $X \in \mathbb{R}^{B \times T \times C \times H \times W}$ with:

- B: Batch Size
- T: Time steps ($T = 5$ frames)
- H, W: Spatial resolution (64×64)
- C: Input Channels (Total 5):
  - Channels 1-3: Thermal data (RGB duplicated or colormap mapped).
  - Channels 4-5: Dense Optical Flow (u, v) computed between consecutive frames to capture heat convection dynamics.

B. Model Architectures

1) Standard Models (Scalar Regression): These models predict a vector of 4 scalar temperature values corresponding to sensor locations.

- **SimpleResNet**: A frame-based ResNet-18 adapted for 5-channel input. Predicts 4 values per frame: $Y \in \mathbb{R}^{B \times 4}$.
- **CNN-LSTM**: Combines a ResNet encoder with an LSTM temporal aggregator. The LSTM processes the sequence of features and outputs the final temperature state: $Y \in \mathbb{R}^{B \times 4}$.

2) Bayesian Models (Uncertainty Quantification): To quantify predictive uncertainty, we employ Variational Inference.

- **Bayesian Neural Networks**: Standard dense layers in the regression head are replaced with ‘BayesLinear‘ layers (distributions over weights).
- **Output**: Returns a tuple $(\hat{Y}, D_{KL})$, where $\hat{Y}$ is the prediction and D_{KL} is the Kullback-Leibler divergence regularization term.
- **Inference**: Uncertainty is estimated via Monte Carlo sampling (multiple forward passes).

3) Dense Prediction and Physics-Informed Models:

- **U-Net**: An encoder-decoder architecture predicting a full spatial temperature map $\hat{T}_{map} \in \mathbb{R}^{B \times 1 \times 64 \times 64}$.
- **Physics-Informed Loss (Bioheat)**: Models are trained with a composite loss function:

  $$\mathcal{L} = \mathcal{L}_{MSE} + \lambda_{phys} \mathcal{L}_{bioheat}$$

  where $\mathcal{L}_{bioheat}$ enforces the Pennes’ Bioheat equation:

  $$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) - \omega_b \rho_b c_b (T - T_a) + q_m + \rho c \mathbf{v} \cdot \nabla T$$

  Here, the convection term $\mathbf{v} \cdot \nabla T$ utilizes the Optical Flow input channels.